Geophys. J. Int. (2005) **163,** 501–517 doi: 10.1111/j.1365-246X.2005.02741.x

Palaeomagnetism and $^{40}Ar/^{39}Ar$ geochronology from the South Taimyr igneous complex, Arctic Russia: a Middle–Late Triassic magmatic pulse after Siberian flood-basalt volcanism

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Accepted 2005 July 6. Received 2005 June 17; in original form 2004 November 11

SUMMARY

New palaeomagnetic results and $^{40}Ar/^{39}Ar$ ages from mafic sills and extrusive rocks from the South Taimyr igneous complex (75°N, 100°E) document late Middle–Early Late Triassic igneous activity in Arctic Siberia. The palaeomagnetic pole determined from the sills (47°N, 122°E, $dp/dm = 5/5$) plots on the 230–220 Ma portion of the apparent polar wander path for Euramerica, and is statistically different from the mean 250 Ma Siberian Traps pole. $^{40}Ar/^{39}Ar$ ages from three of the sills yield crystallization ages of *ca*. 229–227 Ma and confirm the pole age. The Taimyr igneous rocks are folded together with Carboniferous to Lower Triassic continental sedimentary rocks of the northern Siberia margin and are unconformably overlain by Early Jurassic sedimentary units that place an upper limit on the cessation of the Late Triassic folding event. In contrast to the sills, the palaeomagnetic pole obtained from the extrusive volcanic rocks (59°N, 146°E, $dp/dm = 14/15$) overlaps within uncertainty with previous Siberian Traps results, but is consistent with any age between 250 and 220 Ma. The $^{40}Ar/^{39}Ar$ signatures of the extrusive rocks were largely very disturbed and only one sample yielded a precise and interpretable age of 248 Ma.

The intrusive rocks in the South Taimyr igneous complex post-date eruption of the Siberian Traps flood basalts by about 20–25 Myr and are contemporaneous with reported ages from granitic rocks on Novaya Zemlya and late intrusive and pyroclastic rocks from the margins of the Siberian Traps igneous province. The extrusive rocks in South Taimyr may be age-correlative with the extensive Siberian Traps flood basalts with which they have previously been linked; however, our data alone cannot confirm the comagmatic nature of the extrusive rocks within South Taimyr or their genetic link, if any, to the igneous activity that produced the Siberian Traps flood basalts.

Key words: geochronology, igneous rock, palaeomagnetism, Siberian Traps, Taimyr, Triassic.

INTRODUCTION

The Taimyr peninsula lies between the Kara Sea and the Laptev Sea, and contains igneous, metamorphic and sedimentary rocks of Proterozoic to Cretaceous ages. Information from Taimyr is important for understanding the tectonic evolution of a much wider area of Arctic Eurasia, but lack of critical data and contradictory interpretations have compromised this understanding: even the ages of major continental collision events are controversial. From north to south, the peninsula comprises accreted crystalline units of Pre-Cambrian age, the fold-and-thrust Vendian–Carboniferous margin sequence of the northern (present coordinates) Siberia craton, and a folded sequence of Carboniferous to Lower Triassic, continental clastic rocks with interleaved Permo-Triassic rocks of the Taimyr igneous complex (Fig. 1) (Zonenshain *et al.* 1990; Vernikovsky 1997; Vernikovsky *et al.* 2003). Flat-lying Jura–Cretaceous rocks unconformably overlie these units in the southern part of the Taimyr peninsula. Together, these units preserve a Late Palaeozoic to mid-Mesozoic history defining amalgamation related to the collision of Siberia and Euramerica as Pangea was assembled, and the onset of rifting that signalled northern Pangea break-up.

New, high-quality isotope ages from the northern Taimyr crystalline units have resolved the timing of some key Pre-Cambrian and Permo-Carboniferous events related to that part of the peninsula

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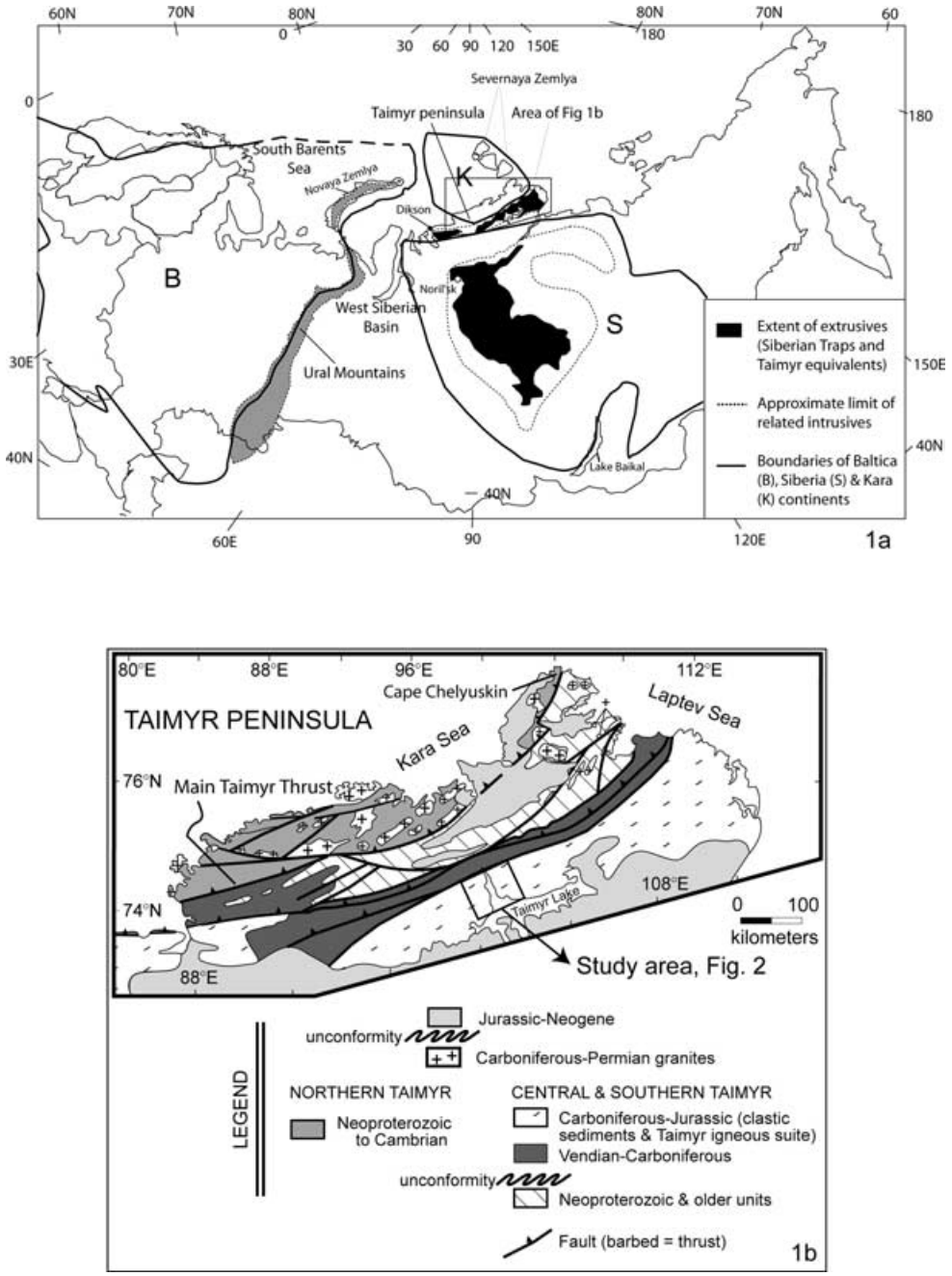


Figure 1. (a) Map showing location of the Taimyr Peninsula in Arctic Siberia. Outlines are shown for the principal continents of palaeogeographic significance for Taimyr in the Palaeozoic and Early Mesozoic: B = Baltica; S = Siberia, K = Kara. (b) Simplified geological map of the eastern Taimyr Peninsula. The study region is identified on the northern margin of Taimyr Lake. Modified from Torsvik & Andersen (2002). See also Fig. 2.

(Vernikovsky 1997; Pease & Vernikovsky 1998). It is generally agreed that northern Taimyr was part of the Kara Block from the latest Pre-Cambrian to Early Palaeozoic time, separated from Siberia by the Uralian Ocean. However, the precise timing, duration and kinematics of collision between the Kara Block and the northern Siberia passive margin as the Uralian Ocean was closed are not well constrained. Kara–Siberia collision is suggested to have occurred in the Permo-Carboniferous, in the Jurassic–Cretaceous, or as part of a continuous process from Permo-Carboniferous to Early Jurassic time (Khain 1985; Zonenshain *et al.* 1990; Otto & Bailey 1995; Vernikovsky 1997; Inger *et al.* 1999; Torsvik & Andersen 2002). Lack of consensus on the timing, duration and kinematics of Kara–Siberia collision affects evaluation of both Late Palaeozoic 'Uralian-age' tectonics, suggested to link Taimyr with Novaya Zemlya and the Polar Urals (Otto & Bailey 1995; Pease & Vernikovsky 1998) (Fig. 1a), and understanding of the cause(s), magnitude and regional significance of Early Mesozoic deformation documented from Timan-Pechora, the eastern Barents Sea, to Novaya Zemlya, the West Siberian Basin and Yenisey-Khatanga Trough, and into Taimyr (Kulikov *et al.* 1972; Khain 1985; Korago *et al.* 1989; Sobornov 1994; Otto & Bailey 1995).

The South Taimyr igneous complex is suggested to be contemporaneous, if not comagmatic, with the Siberian Traps; therefore, precise isotopic ages for the Taimyr igneous rocks would provide

information on what has been suggested to be the northernmost exposure of the voluminous Siberian Traps igneous province. However, few modern isotopic ages exist from South Taimyr and none from the mafic basalt flow and sill sequences. The 'Permo-Triassic' age heretofore assigned to the South Taimyr igneous complex is based on lithostratigraphic, biostratigraphic, geochemical and palaeomagnetic correlations between Taimyr and the well-dated Siberian Traps (*ca*. 251 Ma; Kulikov *et al.* 1972; Zolotukhin & Al'mukhamedov 1988; Renne & Basu 1991; Dalrymple *et al.* 1995; Gurevitch *et al.* 1995; Reichow *et al.* 2002). Since the rocks of the Taimyr mafic igneous suite are folded together with the Carboniferous–Lower Triassic continental clastic succession that they intrude, precise ages for the time of their extrusion and emplacement would also provide a maximum age for the onset of Mesozoic, fold-related deformation in South Taimyr, and would give a stronger basis for evaluating processes responsible for genesis of large volumes of magmatic rocks in Siberia in Permo-Triassic times.

We conducted a palaeomagnetic and $^{40}Ar/^{39}Ar$ study of the mafic igneous rocks in southeastern Taimyr along Taimyr Lake and the southern section of the Taimyr River (Figs 1 and 2) to address the age of the Taimyr igneous units. We report herein palaeomagnetic data from a suite of flows, sills and dykes from the Taimyr igneous suite, as well as $^{40}Ar/^{39}Ar$ furnace step-heating data from plagioclases and whole rocks yielding precise ages from three sills and one flow.

REGIONAL SETTING

The Taimyr Peninsula has generally been divided in previous studies into three zones: North, Central and South Taimyr. North Taimyr comprises Neoproterozoic metasediments of the Kara Block, metamorphosed up to amphibolite facies, with numerous intrusions, some of which yield Permo-Carboniferous ages (Fig. 1) (Vernikovsky 1997; Pease & Vernikovsky 1998). Central Taimyr incorporates various accreted Pre-Cambrian crystalline units including gneisses and island-arc and ophiolite fragments; Permo-Carboniferous intrusive rocks are also prevalent in this zone (Vernikovsky 1997; Pease & Vernikovsky 1998; Pease *et al.* 2001). The Pre-Cambrian accreted terranes of Central Taimyr are unconformably overlain by the Riphean to Middle–Carboniferous continental margin succession representing the stable platform sequence of the northern Siberia passive margin, including South Taimyr. In South Taimyr a fault contact to the south of the platform sequence juxtaposes a package of Upper Carboniferous to Lower Triassic shallow-marine and continental clastic rocks

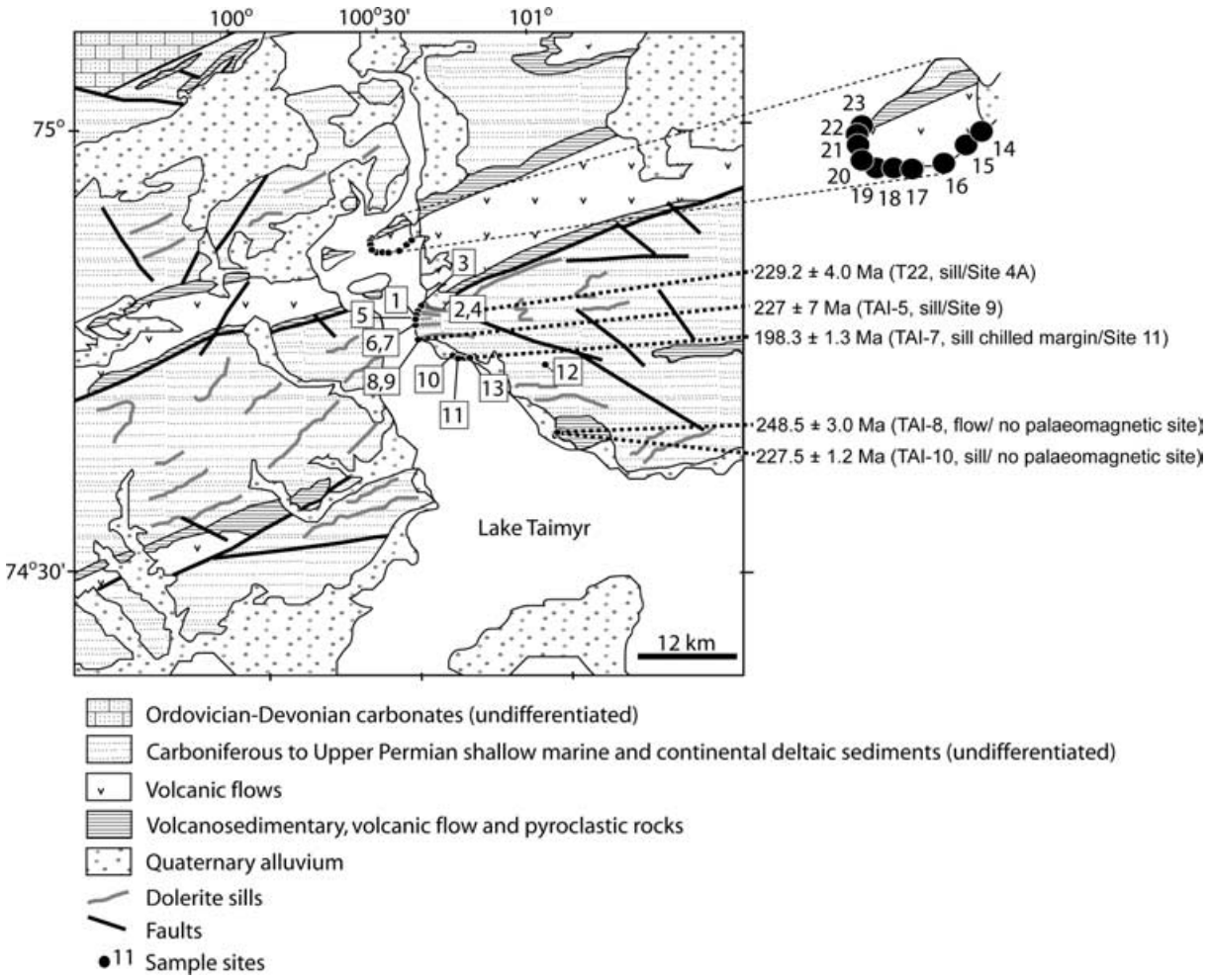


Figure 2. Geological map after Bezzubtsev *et al.* (1983). Palaeomagnetic and $^{40}Ar/^{39}Ar$ sampling sites are indicated with black dots (site numbers in boxes and labelled 1, 2, etc.). An expanded area around Hoffman peninsula showing palaeomagnetic sites 14–23 is indicated to the right of the main map. $^{40}Ar/^{39}Ar$ ages and associated sample numbers and palaeomagnetic sites for four sills and one flow are also indicated to the right of the map. Ages are cited at 1σ. Palaeomagnetic samples from the study of Torsvik & Andersen (2002) were collected northward along Taimyr River, beginning in the Palaeozoic sequences at the northwest edge of this map.

interleaved with the 'Permo-Triassic' extrusive and intrusive rocks of the Taimyr igneous suite (Figs 1 and 2) (Zonenshain *et al.* 1990; Vernikovsky 1997; Pease & Vernikovsky 1998; Vernikovsky *et al.* 2003).

The predominant structural grain throughout the three zones is oriented NE–SW with fold axes and SE-verging regional thrust faults following this trend (Fig. 1b). The Main Taimyr Thrust separates Central and North Taimyr (Fig. 1b). The main orientation of the structural features in North Taimyr relative to Central Taimyr and the cross cutting relationships between NE–SW-striking thrusts and Late Carboniferous–Permian intrusive rocks (Fig. 1b) were used to interpret an 'Uralian' age for amalgamation/collision of the North Taimyr block with Central Taimyr (northern Siberia) (Zonenshain *et al.* 1990; Vernikovsky 1997; Pease & Vernikovsky 1998; Inger *et al.* 1999). The various amalgamation scenarios for Taimyr differ mainly in the timing proposed for Central and South Taimyr collision. Vernikovsky (1997) and Inger *et al.* (1999) advocated Riphean accretion of Central and South Taimyr against the northern Siberian margin, with constraints provided by the Vendian–Carboniferous platform and slope sedimentary package that unconformably overlies Central Taimyr crystalline terranes. In contrast, Zonenshain *et al.* (1990) proposed that the collision of North and Central Taimyr occurred in Permo-Carboniferous times, with the collision of the united North-Central Taimyr entity against the northern Siberia passive margin (South Taimyr) in Jurassic–Cretaceous times. In the Zonenshain *et al.* (1990) model, a suture (thrust?) is required between Central and South Taimyr. The evidence for this suture has been questioned (Vernikovsky 1997; Inger *et al.* 1999; Torsvik & Andersen 2002) and we follow the structural evidence presented by Inger *et al.* (1999) that supports a single Central-South Taimyr entity that was part of Siberia since the Riphean.

In South Taimyr, three key age constraints on the timing of igneous activity and deformation are afforded by the folded, Palaeozoic–Mesozoic stratigraphy:

(1) the entire Palaeozoic carbonate-dominated platform sedimentary package (South Taimyr) contains no obvious major angular unconformities (Inger *et al.* 1999);

(2) the Upper Carboniferous to Lower Triassic continental clastic rocks of South Taimyr signify change from stable carbonate platform deposition to the active influx of clastic detritus, possibly linked to unroofing of the northern Siberian margin to the south or to erosion of the developing Uralian collision belt to the north and west (Inger *et al.* 1999) and

(3) to the south of the folded Late Palaeozoic– Triassic sedimentary and igneous package in South Taimyr, weakly lithified, flat-lying Hettangian–lower Pliensbachian (Lower Jurassic) strata of the Zimnaya Formation lie on an eroded surface of Triassic and Palaeozoic rocks in the Yenisey-Khatanga Trough (Krymgol'ts 1976; Shurygin 1978) (Figs 1 and 2). These indubitable stratigraphic limits provide an Early Triassic to earliest Jurassic bracket for the youngest compressional deformation episode in the Taimyr Peninsula responsible for the strong NE–SW structural grain observed today. A structural interpretation of the study area in southeastern Taimyr was presented in Inger *et al.* (1999). Palaeomagnetic results from the Palaeozoic passive margin sedimentary sequence north of our sampling area were presented in Torsvik & Andersen (2002).

SAMPLING AREA

According to map evidence (Bezzubtsev *et al.* 1983), the Upper Carboniferous to Upper Permian, predominantly terrigenous succession intercalates with the mafic flows, dykes and pyroclastic rocks of the Taimyr igneous complex in South Taimyr. No Lower Triassic clastic sediments are apparently present in the immediate area we studied, although Upper Permian to Lower Triassic clastic rocks are described elsewhere in South Taimyr (Egorov & Kulikova 1989; Mogucheva & Betekhtina 1998). The eruptive units are exposed primarily within a large band that extends across the length of the Taimyr Peninsula and are flanked to the north and south by folded, Late Palaeozoic clastic rocks impregnated with dolerite sills (Figs 1 and 2). The igneous rock types include dolerite sills, mafic dykes, basaltic flows and a variety of pyroclastic and volcanosedimentary rocks. The sills are 3–5 m thick and intrude deltaic sand packages with some limey siltstone intercalations; baked contacts occur at interfaces between sediments and sills. The massive basaltic flows are 3–5 m thick with well-preserved columnar jointing. The pyroclastic and volcanosedimentary rocks range from massive, amorphous, subaerial, vesicular flows to subaerial–subaqueous pyroclastic rocks comprising alternating units of flattened pillow basalts, pahoehoe flows, thin (mm-scale) ash layers, laharic breccia deposits and volcanosedimentary rocks. On the 1:500 000 Taimyr geological map (Bezzubtsev *et al.* 1983), the eruptive and intrusive rocks are generally given Permo-Triassic age designations. With new palaeomagnetic and $^{40}Ar/^{39}Ar$ ages described below, these age designations can now be revised.

SAMPLE DESCRIPTION

A total of 23 sites were sampled for palaeomagnetic and $^{40}Ar/^{39}Ar$ analysis (Tables 1–3) including 11 dolerite sills, eight basalt flows, one mafic dyke and one pyroclastic flow. The remaining two sites were sampled in baked sediments adjacent to two of the sills, and were intended for palaeomagnetic contact tests. A mixture of field coring (25 mm cores) and hand sampling was employed, with orientation performed by sun compass whenever possible, as well as by magnetic compass. All the sites in eruptive rocks have fairly consistent tectonic dips of around 30° SSE. Sill orientations show more variation, with one steeply dipping site (62°) and the remainder ranging between 8° and 28°, thus making a palaeomagnetic fold test feasible. From the 23 sites, 11 samples were selected for $^{40}Ar/^{39}Ar$ analysis (Fig. 2).

All samples had the primary mineral assemblage plagioclase, clinopyroxene, olivine and oxides. The dolerite sills generally have a medium-grained, ophitic to subophitic texture; olivine occurs in both cumulophyric arrangements and as single crystals scattered throughout the rock. The sill from within the vesicular flow unit in the southeastern part of the field area has an equigranular texture with scattered cumulophyric plagioclase. The flows have equigranular, intersertal to hyalo-ophitic textures where spaces between individual, fine-grained, plagioclase laths may be filled with clinopyroxene, olivine, oxides and glass.

In all samples, incipient, but mild, post-intrusive alteration is evident in the form of iddingsite and/or rare chlorite on olivine; rare, patchy sericite on plagioclase is also observed although the samples contained dominantly clean, unaltered feldspar phenocrysts. The magnetic mineralogy of the samples is described together with rock magnetic properties below.

PALAEOMAGNETIC RESULTS

Magnetic remanence was measured on a JR5A spinner magnetometer. A total of 178 specimens from the 23 sites were demagnetized, using both alternating field (2G demagnetizer) and thermal

Table 1. Statistics from Taimyr sill and sediment sites (mean location 74.8°N and 100.6°E). J = mean NRM intensity; N = number of demagnetized samples; n = number of samples/sites used to calculate means. $D(^\circ)/I(^\circ)$ = mean declination/Inclination; k = Fisher precision parameter; α_{95} = 95 per cent confidence circle of mean; BDEC(°) /BINC(°) = Bedding corrected mean declination/inclination; VGP = virtual geomagnetic pole; dp/dm = semi-axes of 95 per cent confidence around the pole. No means were calculated from sediment sites 2 and 7 owing to poorly defined components.

Site	Rock type	J(A m^{-1})	n	N	D(°)	I(°)	k	α_{95}	BDEC(°)	BINC(°)
1	Sill	0.55	7	10	121	75	328	3.3	150	72
2	Limestone	0.003	0	6	—	—	—	—	—	—
3	Sill	0.37	9	10	119	63	174	3.0	155	80
4	Sill	0.35	6	8	128	68	455	3.1	146	70
5	Sill	0.17	6	7	93	83	430	3.2	167	81
6	Sill	0.16	6	6	60	78	369	3.5	144	73
7	Baked sed.	0.001	0	4	—	—	—	—	—	—
8	Sill	0.56	8	8	25	75	305	3.2	147	73
9	Sill	1.06	8	9	64	83	129	4.9	159	72
10	Sill	0.41	8	11	79	77	209	3.8	137	73
11	Sill	0.31	7	10	108	74	107	5.9	155	66
12	Sill	0.67	9	9	11	34	121	4.7	132	80
13	Sill	0.52	8	10	36	74	390	2.8	159	75
Site mean			11	13	70	77	241*	2.9*	150	74
VGP: 47.1°N, 121.6°E, dp/dm = 4.8/5.3										

*k and α_{95} calculated from bedding corrected data.

Table 2. Statistics from Taimyr basalt sites (mean location 74.9°N and 100.5°E). See Table 1 for legend.

Site	Rock type	J(A m^{-1})	n	N	D(°)	I(°)	k	α_{95}	BDEC(°)	BINC(°)
14	Basalt flow	0.16	7	7	2	69	237	3.9	147	74
15	Basalt flow	2.67	4	7	7	61	354	4.9	124	79
16	Basalt flow	1.24	6	8	357	61	509	3.0	94	78
17	Basalt flow	2.05	8	8	19	61	1132	1.6	84	57
18	Dyke	1.82	8	8	198	−71	318	3.1	250	−74
19	Basalt flow	1.59	7	7	11	64	1220	1.7	102	76
20	Basalt flow	0.78	8	8	9	61	596	2.3	82	76
21	Pahoehoe	0.005	5	7	4	68	830	2.7	118	81
22	Pyroclastic	0.16	3	3	300	86	461	5.8	155	70
23	Basalt flow	0.48	7	7	309	57	374	3.1	256	78
Site mean*			10	10	0	68	40*	7.8*	109	78
VGP: 59.3°N, 145.8°E, dp/dm = 13.9/14.7										

*Site 18 inverted; k and α_{95} calculated from bedding corrected data.

(MMTD1 furnace) demagnetization. Although both methods yielded similar components, thermal demagnetization was preferred for the majority of the specimens (151), partly because alternating field demagnetization gave rise to spurious components attributed to Gyroremanent magnetization (GRM) at some sites (Stephenson 1981). Characteristic remanence components were calculated using principal component analysis (Kirschvink 1980).

Mean intensities of natural remanent magnetization (NRM) for all sites are given in Tables 1 and 2. In general, remanence quality is excellent (Figs 3 and 4), excepting the two sediment sites

Table 3. Summary table of ^{40}Ar/^{39}Ar data.

Sample	Mineral	Rock type	WMA (Ma) $\pm1\sigma$	Steps used/ (per cent ^{39}Ar)	Inverse isochron (Ma) $\pm1\sigma$	^{40}Ar/^{36}Ar intercept	SUMS/ $(n-2)$	Preferred age (Ma) $\pm2\sigma$**
TAI-10	Plagioclase	Dolerite sill (intrudes TAI-8)	227.81 ±0.02	4–12/ 88.5	227.47 ±1.24	302.4 ±19.3	7.6	227.5 ±2.5
T22 (site 4A)	Plagioclase	Dolerite sill	Not calculated	—	229.21 ±4.02	325.4 ±5.1	5.6	229.2 ±8.0
TAI-5 (site 9)	Plagioclase	Dolerite sill	Not calculated	—	226.72 ±7.09	353.7 ±8.9	20.1	227 ±14
TAI-7 (site 11)	Plagioclase	Dolerite sill (chilled margin)	Not calculated	—	198.30 ±1.34	338.7 ±5.1	7.8	198.3 ±2.6
TAI–8	Plagioclase	Basalt flow (intruded by TAI-10)	Not calculated	—	248.51 ±2.97	322.7 ±2.8	1.8	248.5 ±6.0

WMA = weighted mean age calculated from the release spectrum with the steps used as indicated in the adjacent column. All ages except last column cited at 1σ uncertainty without uncertainty in J-value (all samples were irradiated in the same package). Sample names and palaeomagnetic sampling sites are noted in the first column. TAI-10 and TAI-8 were not amenable for palaeomagnetic sampling.

**The preferred crystallization age (the isochron age in all cases) is noted at 2σ uncertainty, including the uncertainty in J-value (1 per cent).

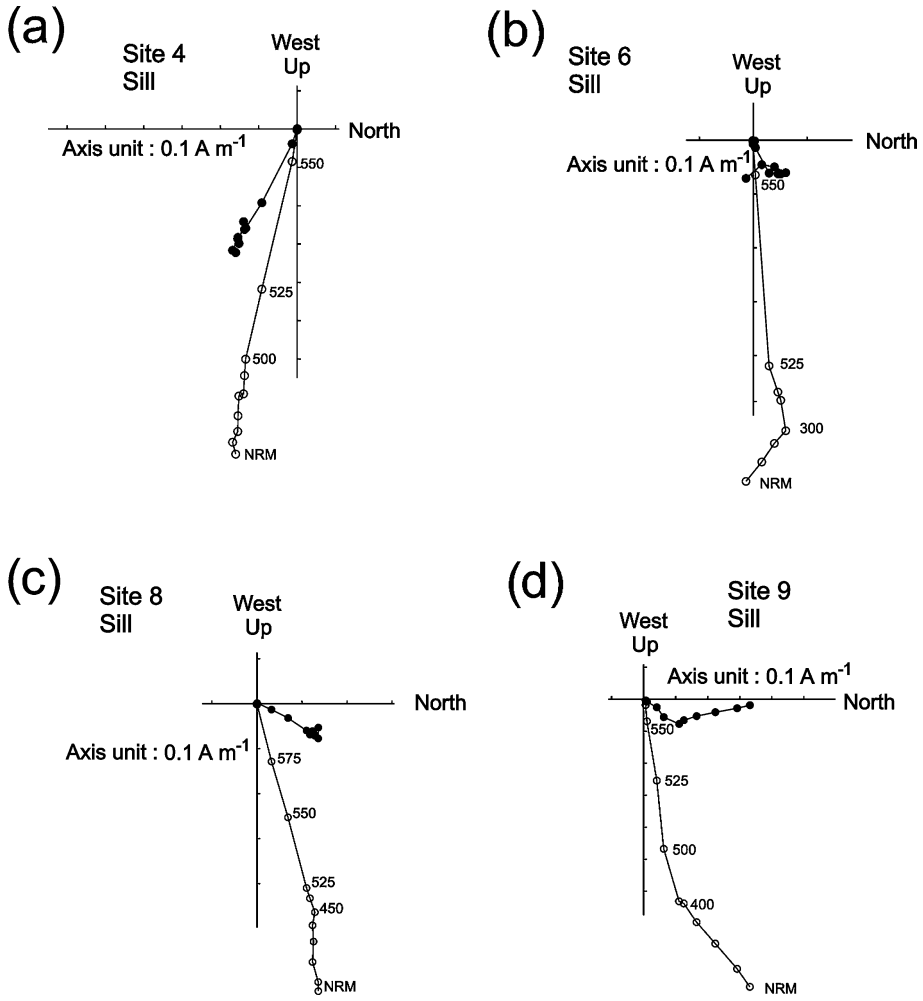


Figure 3. Examples of thermal demagnetization results from Taimyr sills shown in orthogonal vector plots (*in situ* coordinates). Open (solid) symbols represent projections onto the vertical (horizontal) planes respectively.

(2 and 7) that did not yield consistent directions, and will not be discussed further. The 11 sill sites yield steeply down-pointing characteristic remanence components (Fig. 3) with the main unblocking range between 500°C and 575°C. Secondary lower stability components are present in many of the sill specimens (Figs 3b–d), but do not yield consistent directions and were not investigated further.

All basalt flows (Figs 4a–c) yield single-component, down-pointing magnetizations with a more variable blocking range than in the sills. Remanence directions in some specimens remain stable up to maximum blocking temperatures close to 680°C (Fig. 4a) showing that haematite must be present. The only site to carry a reversed magnetization is the dyke, where all demagnetized specimens show well-defined up-pointing characteristic components, often accompanied by a normal polarity overprint (Fig. 4d).

Statistical treatment was performed separately on the sills (sites 1–13) and the remaining volcanic rocks (sites 14–23). This decision is based on the differences in magnetic mineralogy and remanence acquisition time explained below, as well as a possible age difference. Site mean directions for the sills are shown in Table 1. Within-site grouping is good, with Fisher precision parameter k ranging between 107 and 455 for individual sites. Bedding correction results in a substantial improvement in between-site grouping, as can be readily seen in Fig. 5. The data easily pass the classical fold test of McElhinny (1964), identifying the magnetization as pre-tectonic. McFadden & Jones' (1981) more modern fold test is not applicable, since it requires the sites to be grouped on separate 'fold limbs', which is not the case here. The tight grouping of the data after unfolding (Fig. 5b) implies that all the sills were completely flat lying during magnetization, indicating that all the fold deformation must have taken place after cessation of the igneous activity. A very well-defined, bedding-corrected grand mean ($D = 150°, I = 74°, \alpha_{95} = 2.9°$) yields a palaeomagnetic pole position of 47.1°N, 121.6°E ($dp/dm = 4.8/5.3$).

Site mean directions for the volcanic rocks are detailed in Table 2 and Fig. 6. Within-site grouping is excellent, with k-values exceeding 1000 at two sites. All sites exhibit similar strikes and dips, rendering any attempts at a fold test inconclusive, as may be seen in Fig. 6. However, the presence of the reversely magnetized dyke, a high-stability characteristic remanence and magneto-mineralogic considerations (discussed below) all argue in favour of a primary magnetization. The site means are more dispersed than for the sills, yielding a bedding-corrected grand mean of $D = 109°, I = 78°$, $\alpha_{95} = 7.8°$ and associated pole position of 59.3°N, 145.8°E ($dp/dm = 13.9/14.7$).

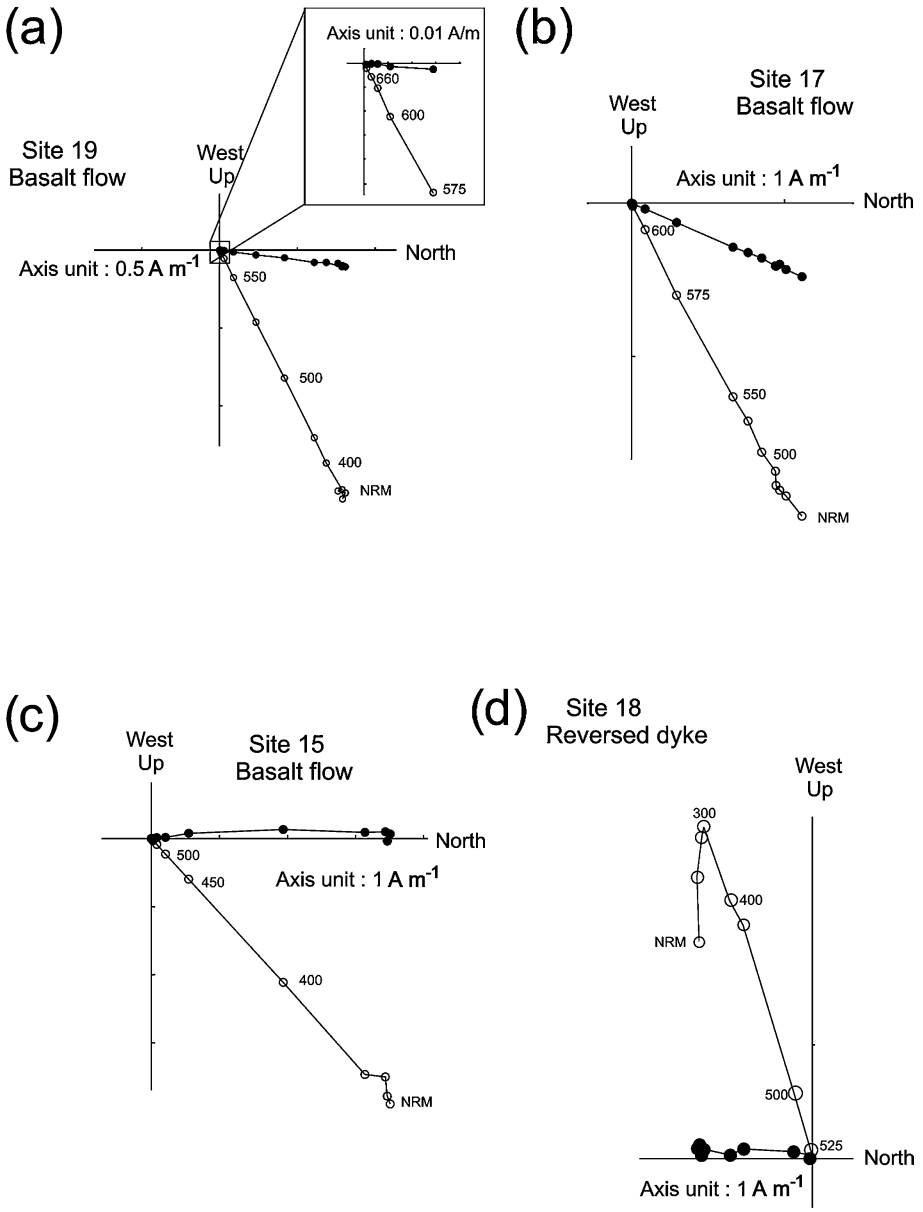


Figure 4. Examples of thermal demagnetization results from basalt flows (a–c) and the mafic dyke (d). Inset in (a) shows enlargement of the final demagnetization steps. Conventions as in Fig. 3.

ROCK MAGNETIC PROPERTIES

Various rock magnetic methods were employed in an attempt to elucidate the timing and stability of the stable remanence, including thermomagnetic curves, coercivity spectrum analysis, optical microscopy and hysteresis curves. We also measured the anisotropy of magnetic susceptibility (AMS) in order to test for potential remanence deflections (Cogne 1987). Mean anisotropy values for all sites are below 5 per cent, suggesting that remanence deflections are not significant, so the AMS results are not further elaborated.

Reflected-light microscopy reveals titanomagnetite as the dominant magnetic mineral in all the volcanic rocks. The sills (Figs 7a and b) contain relatively large grains with ilmenite lamellae (Fig. 7b) attesting to relatively slow initial cooling. Secondary, low-temperature alteration ranging from moderate (Fig. 7a) to quite severe (Fig. 7b) is also evident from granulated grain surfaces and some shrinkage cracks, suggesting the presence of cation-deficient mineral phases. In contrast, most of the basalt flows (Fig. 7c) contain homogeneous skeletal titanomagnetite grains, with no visible evidence of phase unmixing, consistent with rapid initial cooling.

A total of 34 thermomagnetic curves were obtained in air on a horizontal translation balance with a heating/cooling cycle of 100 min in a field of 0.7 Tesla. Typical curves for sills and basalt flows are shown in Figs 8(a) and (b), respectively. The sills all show Curie points close to 580°C, suggesting pure magnetite as the primary remanence carrier. Curves are generally reversible, except for a slightly steeper trajectory below 580°C on heating (*cf.* Fig. 8a), which might be explained by the inversion of a small maghemite constituent to haematite. The basalt flows show slightly more variation, but Curie points generally fall between 450°C and 500°C, indicating a significant titanium content in the magnetic phase. One likely contributor to the difference in Curie points between the two rock types is the difference in cooling rate, with the rapidly cooled basalts allowing less time for unmixing of Ti-rich and Ti-poor phases

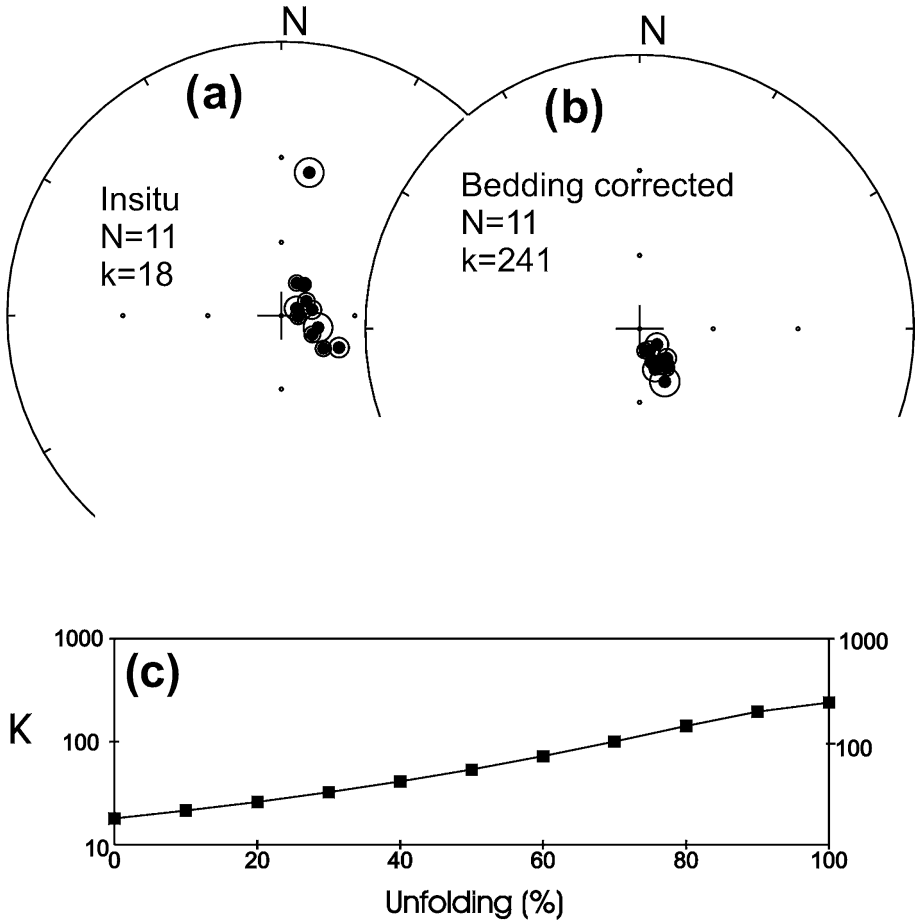


Figure 5. Site mean directions for sills shown with α_{95} confidence circles for individual sites. (a) *In situ*. (b) Bedding corrected. (c) Stepwise fold test showing the variation of Fishers' precision parameter K as a function of percentage unfolding.

in the titanomagnetite grains through high-temperature oxidation, although initial compositional differences may also affect this process.

Isothermal remnant magnetization (IRM) versus field curves were obtained for a total of 25 specimens in maximum fields of up to 4.2 Tesla, using an in-house manufactured solenoid for moderate field strengths (<0.2 T) and a pulse magnetizer for high fields. Several examples for both sills and basalt flows are shown in Fig. 5. Again the sills show very consistent properties (Fig. 9a), saturating in fields slightly below 200 mT, with no evidence of the presence of high-coercivity phases such as haematite. Basalt flows (Fig. 9b) have minimum saturation fields below 100 mT, while some samples combine a rapid initial remanence increase in fields below 300 mT with a small continued increase up to maximum fields of 1.5 Tesla, attesting to the presence of a subordinate haematite phase in addition to the dominant titanomagnetite.

Hysteresis measurements, performed on a Molspin vibrating sample magnetometer, yield J_{rs}/J_s ratios between 0.15 and 0.34 for all 18 specimens measured, placing the magnetic minerals in all types of volcanic rocks in the middle of what is traditionally seen as the pseudo-single domain (PSD) range (Dunlop 1986; Dunlop & Özdemir 1997), although interpretation of this ratio in terms of domain state is somewhat ambiguous when mixtures of different grain sizes and varying titanomagnetite compositions are involved (Tauxe *et al.* 1996; Dunlop 2002).

In summary, the sills have remarkably homogeneous magnetic properties, and seem to contain titanomagnetite grains with almost pure PSD magnetite as the sole magnetic constituent. The presence of ilmenite lamellae attests to relatively slow cooling. The basalt flows have slightly more variable magnetic properties, with PSD titanomagnetite and in some cases subordinate haematite carrying the remanence. Small skeletal grains attest to rapid cooling.

In terms of palaeomagnetic significance, these differences between the two main rock types studied would signify that results from many of the eruptive sites constitute spot readings of the field, whereas the more slowly cooling sills probably allowed enough time for averaging out secular variation. This would explain the extremely small within-site dispersion in the basaltic flows coupled with a much larger between-site dispersion (Table 2), as well as the very consistent site-mean directions for the bedding-corrected sills (Table 1).

^{40}AR/^{39}AR RESULTS

Analysed samples included plagioclase from six dolerite sills and either plagioclase or whole rocks from four basaltic flows and one dyke. Plagioclase from one sill yielded a consistent release pattern from a furnace step-heating experiment that defines a near plateau. The other 10 samples yielded disturbed furnace release spectra attributed to a combination of excess argon and post-emplacement alteration (see below); in four of these cases (three sills and one flow), geologically meaningful ages were obtained from the

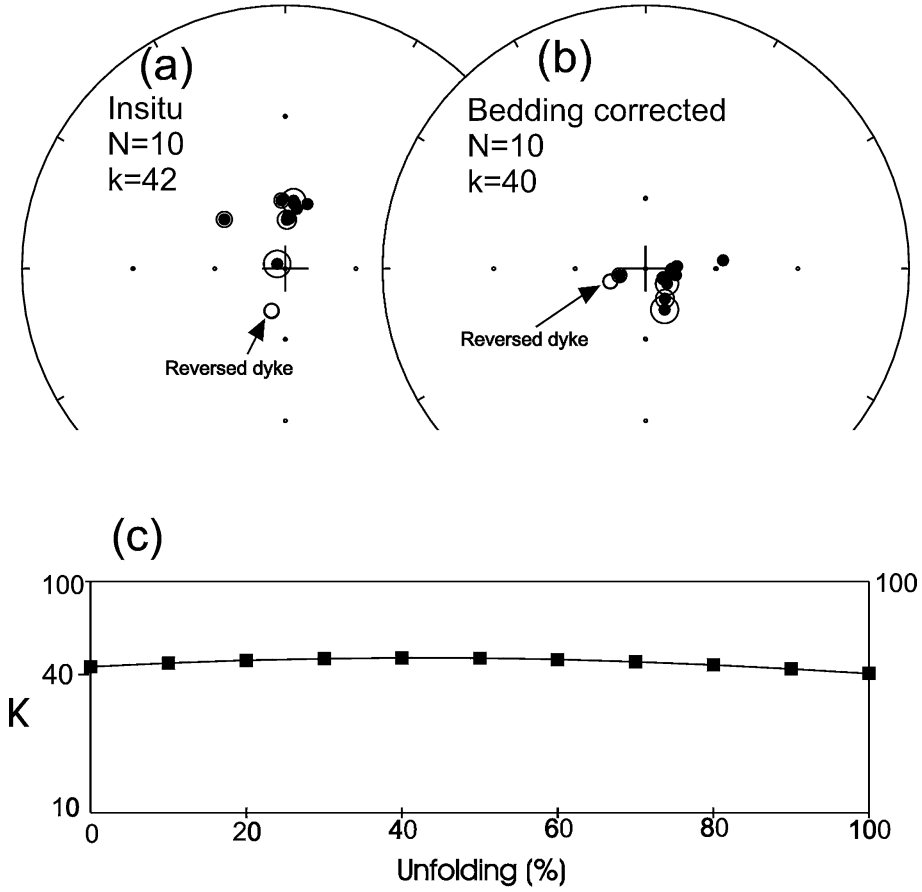


Figure 6. Site mean directions for eruptive rocks. Legend as in Fig. 5.

step-heating data by inverse isochron analysis. The remaining six samples clearly contain several argon reservoirs with complicated degassing signatures not completely related to radiogenic ^{40}Ar derived from decay of inherent ^{40}K; these six samples are not discussed further.

Analytical protocol followed Eide *et al.* (2002) with supplementary information and complete data tables presented in the Appendix. All ages in the descriptive text and figures are cited at 1σ uncertainty, excluding the intra-laboratory uncertainty in J-value since all samples were irradiated in the same package. Ages in the last column of Table 3 are noted at 2σ uncertainty (including J-value uncertainty) to facilitate comparison with age data from other studies in Siberia. Uncertainty on isochron ages was calculated as a standard error. Corresponding palaeomagnetic sites, where appropriate, are listed together with the sample numbers.

SILLS

TAI-10 plagioclase (no palaeomagnetic site)

In the release spectrum, the sample yielded a set of fairly concordant steps defining a near plateau representing 88.5 per cent of the ^{39}Ar gas released in the experiment (Fig. 10). The weighted mean age for this portion of the spectrum, 227.81 ± 0.02 Ma, overlaps within uncertainty of the age determined from the inverse isochron for the same steps (227.47 ± 1.24 Ma). The isochron indicates a trapped (non-radiogenic) argon component of atmospheric value (Table 3). We interpret the isochron age of 227.5 ± 1.2 Ma as the crystallization age for the sill.

T22 plagioclase (palaeomagnetic site 4A)

The high-temperature portion of the disturbed spectrum constitutes *ca*. 50 per cent of the experimental gas released from the sample and yields a range of apparent ages between 219 and 245 Ma; this portion of the spectrum corresponds to the higher K/Ca and lower Cl/K ratios of the experiment (Fig. 11a). The first half of the spectrum comprises four steps with high apparent ages and comparatively lower K/Ca and higher Cl/K ratios. On an inverse isochron diagram these two distinct sections of the release spectrum are resolved into two separated groups that together define a linear array corresponding to an age of 229.21 ± 4.02 Ma with a trapped argon component slightly above atmospheric value (295.5; Table 3). We attribute most of the disturbance observed in the spectrum to this trapped excess argon component (see summary below). Incipient alteration of plagioclase was not evident in thin section and we suggest that the observed changes in K/Ca ratios indicate degassing from pristine, zoned feldspar. We interpret the isochron age of 229.2 ± 4.0 Ma as the age most closely corresponding to crystallization of this sill.

TAI-5 plagioclase (palaeomagnetic site 9)

The release spectrum for plagioclase from this sill has a generally concave-up shape (Fig. 11b); the youngest apparent ages at the base of this 'saddle' are between *ca*. 240 and 230 Ma. The distribution of the data on an inverse isochron diagram shows two different argon reservoirs:

(1) a primary signature comprising the majority of steps, distributed along a linear array (black points on the figure) and

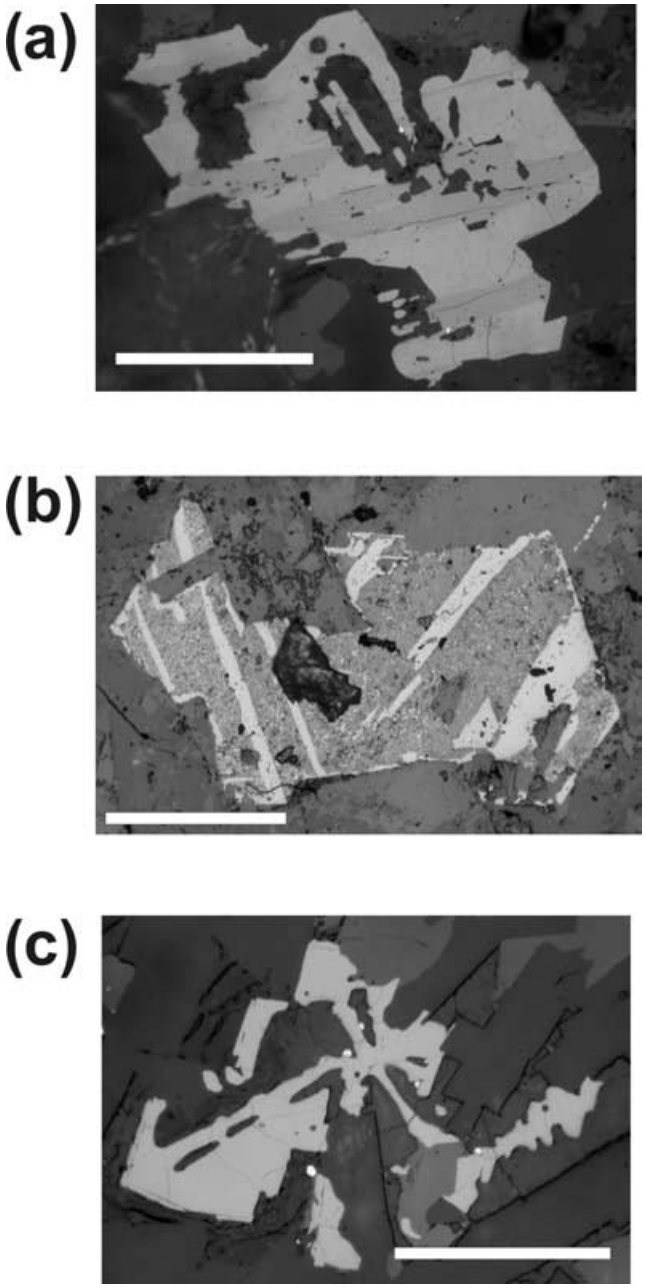


Figure 7. Reflected light micrographs of titanomagnetite grains from sills (a,b) and basalt flow (c). Length of scale bars = 100 μm.

(2) the highest-temperature steps of the experiment that cluster with very little spread off this main data trend (white points on the figure). An isochron fit through the linear array, excluding the highest temperature steps and one aberrant, high-age step in the middle of the experiment, define a line corresponding to an age of 226.72 ± 7.09 Ma with a trapped argon component in excess of atmospheric value (Fig. 11b; Table 3). While some of the variability in the degassing signature, and the 'saddle' shape could be attributed to an excess argon component, the poor correlation of the line to the data implies that the main argon reservoir of the sample is probably more complicated than simple mixing of trapped and radiogenic argon reservoirs. We thus refer subsequently to an imprecise age of

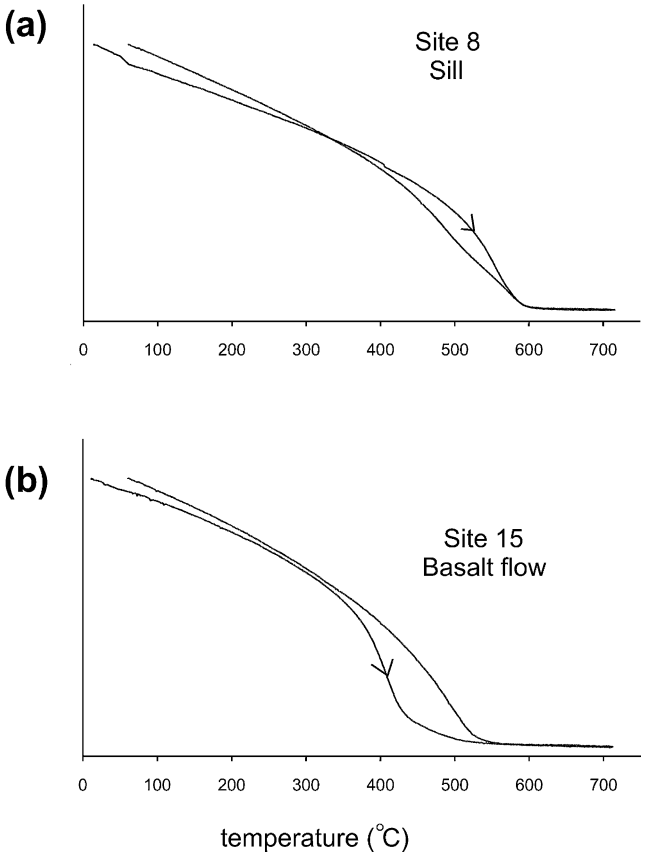


Figure 8. Representative thermomagnetic curves from sill (a) and basalt flow (b) respectively. The arrows identify the heating part of the cycle.

227 ± 7 Ma as most closely approximating the crystallization age of this sill.

TAI-7 (palaeomagnetic site 11)

This sample was from the fine-grained, chilled margin of the sill at this site, in contact with the surrounding sedimentary rocks. The slightly concave-up pattern of the release spectrum for plagioclase is dominated by a set of high-temperature steps with ages between 205 and 199 Ma, representing *ca*. 52 per cent of the ^{39}Ar gas released (Fig. 11c; Table 3). Excluding the first and last steps of the experiment, all of the data on an isochron diagram are distributed along a reasonably linear array, which yields a lower intercept corresponding to an age of 198.30 ± 1.34 Ma with a trapped argon component greater than atmospheric value. Despite the reasonable agreement between the spectrum and isochron ages for this sample, the sample's position at the margin of the sill, adjacent to the water-rich sedimentary rocks that the sill intruded, does not give us confidence to interpret this age as a primary crystallization age for the sill (or sill margin). Rather, the stable palaeomagnetic data from this sill (Tables 1 and 2) and their excellent correspondence to palaeomagnetic data from all the other sills for which we obtain the mid-Triassic pole, suggest that this ∼198 Ma ^{40}Ar/^{39}Ar age probably corresponds to the time of resetting of the argon system along the sill margin. The 198 Ma age should be considered a maximum age for the time of resetting (see also below).

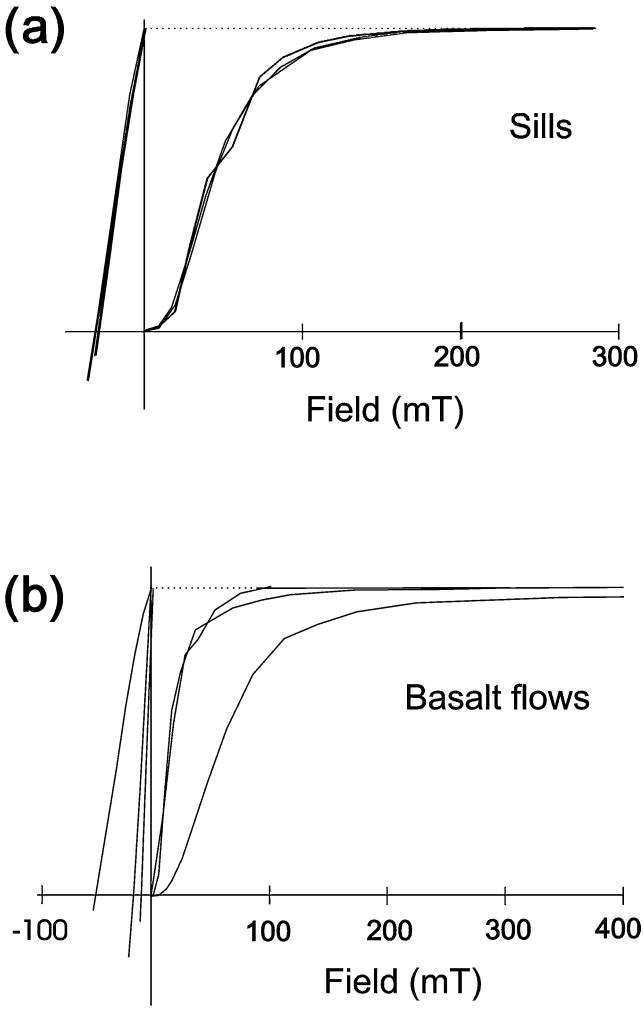


Figure 9. Examples of IRM acquisition and backfield curves for sills (a) and basalt flows (b). Note difference in scale on horizontal axes.

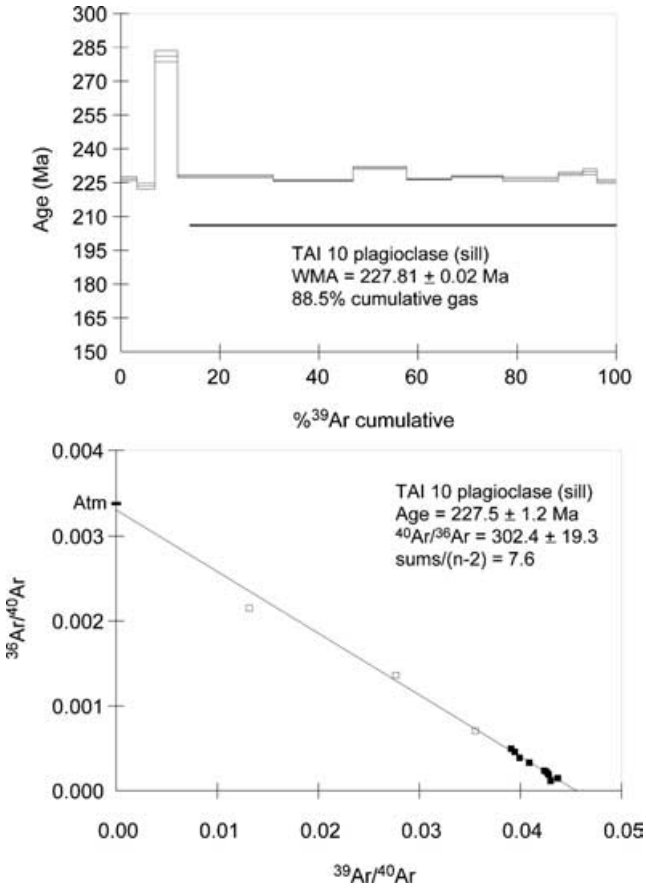


Figure 10. $^{40}Ar/^{39}Ar$ release spectrum and inverse isochron for plagioclase from sill TAI-10. The weighted-mean age from the release spectrum incorporates individual steps weighted according to both amount of ^{39}Ar and uncertainty in the apparent age. In inverse isochron diagrams, poor fit of a line to the data (SUMS/$(n-2) > F$-variate critical value) indicates greater uncertainty than that attributable to analytical error alone. Uncertainties on data points in the isochron are plotted in the figure but are generally smaller than the data point. Black data points are those used in the regression and represent the same steps as those used in the weighted-mean age calculation. Ages are cited at 1σ, excluding J-value uncertainty. See also Table 3.

FLOW

TAI-8 (no palaeomagnetic site; intruded by sill TAI-10)

Pristine plagioclase from this massive, fine-grained, vesicular flow yielded a somewhat irregular release spectrum over most of the experiment (Fig. 11d), although a majority of steps constituting *ca.* 53 per cent of the ^{39}Ar gas released yielded ages between 264 and 246 Ma. On an inverse isochron diagram, all of the data, excluding two anomalous steps in the middle of the experiment, define a well-correlated linear array corresponding to an age of 248.51 ± 2.97 Ma with a trapped argon component slightly above atmospheric value. Incorporating the two slightly anomalous data points in the regression does not alter the isochron age within uncertainty, but yields a poorer fit of a line to the data. The isochron and weighted mean ages are similar within uncertainty and we interpret the isochron age of 248.5 ± 3.0 Ma as the crystallization age of the flow. Appropriately, the intruding sill (TAI-10) has a younger age (227 Ma, see above).

$^{40}Ar/^{39}Ar$ data summary

The crystallization ages for three main sill bodies were all similar within uncertainty at 227.5 ± 1.2 Ma, 229.2 ± 4.0 Ma and 227 ± 7 Ma (Table 3). The age for sill sample TAI-10 is the most precise age from our sample set and indicates no incorporation of any trapped excess argon component. We suggest that intrusion of sill TAI-10 into a dry, sub-aerial basalt flow (TAI-8, from which the older crystallization age of 248.5 ± 3.0 Ma was obtained) facilitated quantitative and uniform retention of radiogenic argon in the feldspars from this sill in an argon reservoir of atmospheric composition. Intrusion into more competent rock may also have buffered the sill from the effects of later regional deformation. In contrast, the other two sills yielded somewhat to very disturbed spectra with trapped argon components above atmospheric value. The disturbances in the argon isotope systems of these two sills are attributed to three interrelated factors:

(1) the sills intruded sediments filled with fluids;
(2) the sills did not cool instantaneously and
(3) the sills and surrounding sedimentary rocks were affected by a deformation event subsequent to intrusion. Magma–fluid interaction resulting from emplacement into sedimentary rocks would have provided a means to disturb the original argon reservoir at the time of crystallization; diffusion during relatively slow cooling could additionally have affected a degree of non-uniform distribution of argon

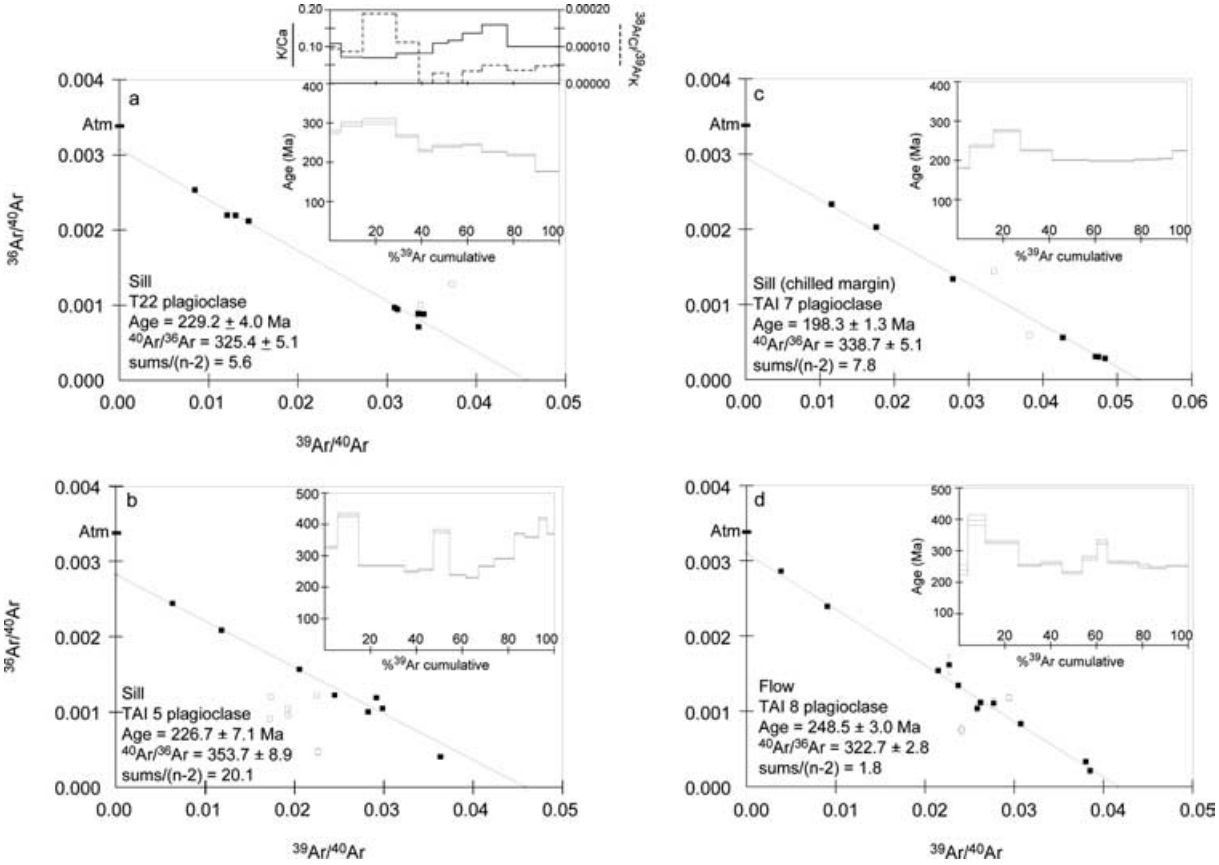


Figure 11. Inverse isochron diagrams with release spectrum insets for plagioclase from three sills T22 (a), TAI-5 (b) TAI-7 (c) and one basaltic flow, TAI-8 (d). Samples T22 and TAI-5 were from the main dolerite sill bodies, while TAI-7 was from a sill chilled margin. K/Ca and Cl/K data also shown for T22. Data presentation otherwise as in Fig. 10.

and potassium in the feldspars that may have been exacerbated by subsequent folding-related deformation. We suggest the effects of the intrusion into water-rich sedimentary rocks affected the dominant, excess-argon signature in the rocks, at the time the rocks were cooling after emplacement. Similar disturbances observed by Dalrymple *et al.* (1995) in $^{40}Ar/^{39}Ar$ plagioclase spectra from intrusive rocks of the Siberian Traps were also attributed to cooling effects. Based on the statistical overlap of the sill ages, regardless the rock type they intruded, and their correspondence with the well-grouped, high-stability palaeomagnetic data that define a pole of the same Mid-Triassic age, we estimate cooling of the sills to have occurred within a maximum of a few hundred thousand years after intrusion.

We interpret the $^{40}Ar/^{39}Ar$ age of *ca*. 198 Ma obtained from the rim of sill TAI-7 as a maximum age for resetting of the argon isotopes along the margin of this rock, and that we attribute to the fold-deformation episode. Field observations indicate that the pronounced competency contrast between the massive sills and the well-bedded alternating units of shale, siltstone and sandstone in the host deltaic succession focused flexural slip (and presumably fluid flow) along the sill contacts during folding. The effects of folding on competent dolerite sills are most pronounced in fold hinge zones where they responded by intense cataclasis accompanied by veining and alteration. We note that a partial resetting age of *ca*. 198 Ma corresponds almost exactly to the Triassic–Jurassic boundary, consistent with stratigraphic constraints on the age of deformation in Taimyr.

DISCUSSION

Fig. 12 shows the palaeomagnetic poles for the sills and eruptive rocks, respectively, compared with the European apparent polar wander path (APWP) of Torsvik *et al.* (2001). For comparison we also show the mean 250 Ma pole for Siberian Traps volcanic rocks (Van der Voo & Torsvik 2004). The sill pole independently confirms the *c*. 229–227 Ma $^{40}Ar/^{39}Ar$ ages obtained from three of the sills. The pole is significantly different from the 250 Ma pole positions for both the Siberian Traps and the European APWP. Comparison with the APWP (Fig. 12) places the magnetization age at 230–220 Ma, again indicating that the sill magmatism is significantly younger than the main Siberian Traps event.

The sill pole is also identical within error limits to the remagnetized sediment pole of Torsvik & Andersen (2002), obtained on sediments of Riphean to Permian ages located approximately 35–50 km NNW of our most northerly sampling sites. Those authors attributed their well-defined pole, from sediments of widely differing ages, to a pervasive regional remagnetization event, which they also placed in the Middle Triassic based on a comparison with the Euramerica APWP; an interpretation that is reinforced by the present study. Such a widespread regional remagnetization, coupled with magmatic intrusions, argues for a significant regional event post-dating the main phase of Siberian Traps volcanism by at least 20 Ma. Furthermore, the positive fold test on the sills confirms that the major folding phase in the area post-dates this regional event. As with the present study, Torsvik & Andersen (2002) also obtained a convincing positive fold test on their sediments, indicating that

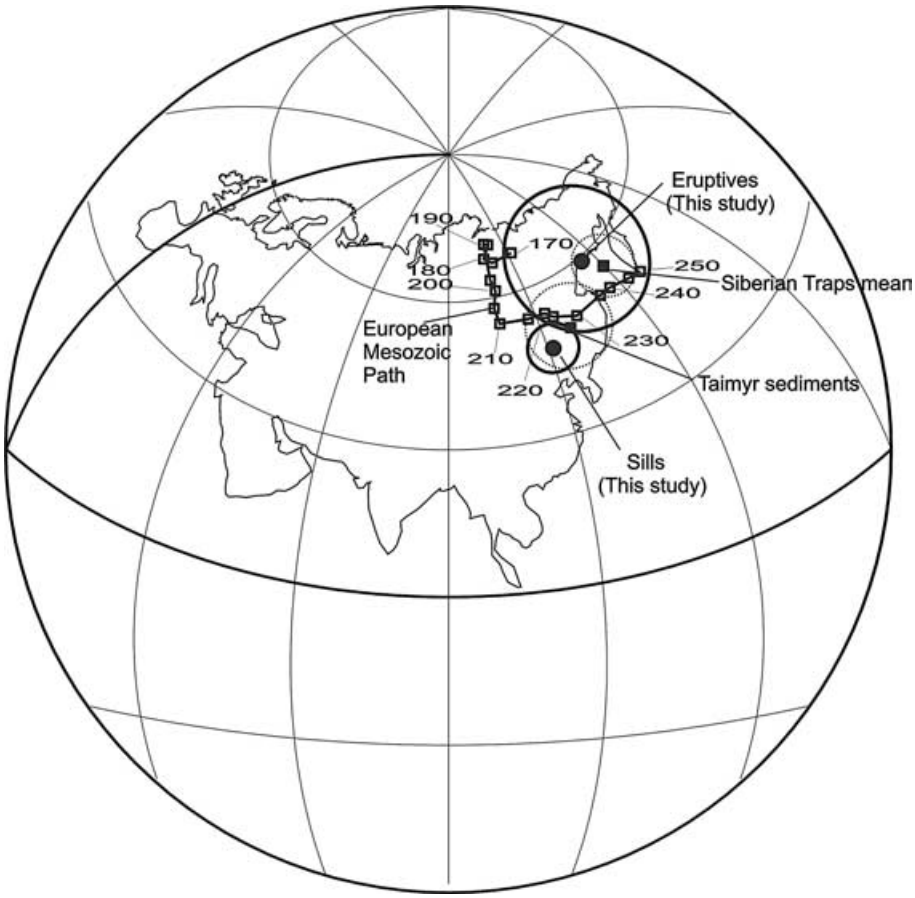


Figure 12. Pole positions for sills and eruptives respectively, shown with dp/dm confidence ovals. Also shown for comparison is the mean pole for Permo-Triassic Siberian Trap volcanics (Van der Voo & Torsvik 2004), the Taimyr sediment pole of Torsvik & Andersen (2002), and the European Mesozoic Apparent Polar Wander Path (Torsvik *et al.* 2001) with numbers indicating ages for points on the path.

their (Mid-Triassic) remagnetization event occurred prior to the onset of folding. Collectively the two studies support the structural and stratigraphic evidence presented in Inger *et al.* (1999), leaving little doubt that the major folding phase in South-Central Taimyr postdates the Mid-Triassic and excludes any direct correlation with the main Late Palaeozoic Uralian deformation.

In contrast to the sill pole, the pole obtained from the volcanic rocks (Fig. 12) falls close to the mean palaeomagnetic pole (Van der Voo & Torsvik 2004) for the Siberian Traps. A recently published and well-documented pole position (56°N, 146°E), from trap volcanics in the Norilsk region *c*. 600 km SW of our study area (Gurevitch *et al.* 2004) lends added credibility to this mean position. The only meaningful $^{40}Ar/^{39}Ar$ age we were able to obtain from a basalt flow (sample TAI8—248.5 Ma) also suggests an age concurrent with the main phase of Siberian Traps volcanism for at least some of the eruptive volcanic rocks of South Taimyr. We note, however, that this age came from a flow at the southern extreme of our study area, well removed from the main belt of extrusive rocks, where all of the palaeomagnetic sampling took place (*cf.* Fig. 2). The error limit on the palaeomagnetic pole from the extrusive rocks would allow for any age between 250 and 220 Ma. On the basis of our data alone, we cannot, therefore, exclude that the main body of eruptives in the Taimyr mafic igneous complex has ages similar to the sills (230–220 Ma), rather than an age similar to the main Siberian Traps volcanism (*ca*. 250 Ma). However, stratigraphic evidence from areas of South Taimyr adjacent to our sampling location implies an Early Triassic age for the eruptives (Egorov & Kulikova 1989).

As the sills of South Taimyr are folded together with the sedimentary units they intrude, the youngest folding deformation cannot be older than 229–227 Ma. Isolated occurrences of weakly consolidated, flat-lying, Toarcian sediments occur in North-Central Taimyr (Dibner & Miroshnikov 1962), and undeformed Hettangian–lower Pliensbachian strata sit above an unconformity in the Yenisey-Khatanga Trough (Krymgol'ts 1976; Shurygin 1978), suggesting deformation ended by earliest Jurassic time (*c*. 195 Ma). These constraints support Inger *et al.* (1999), who suggested that any deformation related to a possible Permo-Carboniferous collision of the Kara block with northern Siberia must have been of a 'thin-skinned' nature (without strong fold development in the passive margin sedimentary sequence), and that the second deformation episode, of latest Triassic–Early Jurassic age, involved more intense folding. We attribute the 198 Ma resetting age obtained from the chilled margin of one eastern Taimyr sill to the latest Triassic–earliest Jurassic deformation in Taimyr.

In this context, it is useful to compare our new $^{40}Ar/^{39}Ar$ ages from the Taimyr igneous suite to isotopic age data from Triassic–Jurassic igneous rocks of the Siberian Traps, Novaya Zemlya and the West Siberian Basin (Fig. 1a). The Siberian Traps are the most reliably dated of the three regions with a large number of modern, high-precision U-Pb and $^{40}Ar/^{39}Ar$ data that confirm a short eruption interval for the Siberian Traps flood basalts at or near the

Permo-Triassic boundary, between *ca*. 251 and 248 Ma (Renne & Basu 1991; Campbell *et al.* 1992; Dalrymple *et al.* 1995; Venkatesan *et al.* 1997; Kamo *et al.* 1996; Czamanske *et al.* 2000; Reichow *et al.* 2002). These ages have formed the basis for stratigraphic correlations between the Siberian Traps and other regions of northern Siberia, including western Taimyr (Gurevitch *et al.* 1995).

In addition to the flood basalts, $^{40}Ar/^{39}Ar$ ages from a group of intrusive and extrusive rocks along the western margin of the Siberian craton near Norilsk (Fig. 1a) have also demonstrated that some of the magmatism in that region continued into the Mid-Triassic (*ca*. 235–223 Ma) (Dalrymple *et al.* 1995). These authors suggested that one Mid-Triassic granodiorite (the Bolgokhtokh granite) is similar (geochemically) to granitic rocks of western Taimyr, recently dated at 249–233 Ma (Vernikovsky *et al.* 2003; see also Czamanske *et al.* 2000). The high-precision, Middle Triassic ages from near Norilsk may vindicate some of the previously cited K–Ar ages whereby Siberian Trap magmatism was suggested to span from *ca*. 248 to 172 Ma (Kulikov *et al.* 1972; Zolotukhin & Al'mukhamedov 1988); these K–Ar ages have been otherwise difficult to evaluate due to lack of data documentation. Adding our new, Mid-Triassic Taimyr sill ages to similarly aged intrusive rocks near Norilsk indicates a sizeable areal extent to magmatism on the Siberian Platform at this time. We note that the occurrence of Mid-Triassic magmatism in Taimyr is also compatible with the track of the Siberian plume predicted by Lawver *et al.* (2002).

If amalgamation of Siberia and Euramerica by the Late Permian coincided with Pangea amalgamation (Torsvik *et al.* 2001), the initiation of the West Siberian Basin serves as a general time marker for the start of extension that signalled the break-up of northern Pangea. Rifting in the West Siberian Basin began in the Late Permian and continued to the Early Triassic, accompanied by rift-related volcanism (Lobkovsky *et al.* 1996); the eruption of the Siberian Traps flood basalts at the Permo-Triassic boundary similarly overlaps with this rifting phase. Mid-Triassic times in the West Siberian Basin were probably dominated by thermal subsidence (Lobkovsky *et al.* 1996) while magmatism apparently continued along the western margin of the Siberian Traps (Dalrymple *et al.* 1995) and in Taimyr (this study). At the Triassic–Jurassic boundary, inversion in Timan-Pechora and the South Barents Sea (Sobornov 1994; Otto & Bailey 1995), folding on Novaya Zemlya (Otto & Bailey 1995) and folding in Taimyr (Inger *et al.* 1999) were contemporaneous with inversion (Nikishin *et al.* 1996) and/or extension (Lobkovsky *et al.* 1996) in the West Siberian Basin. Regional subsidence and tectonic quiescence characterize most of Arctic Siberia from latest Early Jurassic time onwards.

The Late Triassic–Early Jurassic deformation in Arctic Siberia was thus a really extensive, and apparently ended rather abruptly, according to new ages from Taimyr. The geodynamic cause for this regional Late Triassic–Early Jurassic deformation remains unclear, and more work in the region is clearly required before a detailed model for the tectonic event can be presented. Continued effects of SE-thrusting from the Permo-Carboniferous collision of the Kara block with northern Siberia have been suggested (Otto & Bailey 1995), but remain difficult to quantify. The ultimate cause may lie in even more fundamental and far-field effects, indicated by systematic changes in the continental configuration of northern Pangea in Late Triassic times (Torsvik *et al.* 2001). One possibility is that the deformation is related to the initiation of the Arctic Ocean (Inger *et al.* 1999), although further data from both northern Russian and the Arctic Ocean are required to demonstrate such a link.

CONCLUSIONS

Extrusive volcanic rocks and sills in South Taimyr have previously been assumed to constitute the northernmost expression of Permo-Triassic Siberian Traps volcanism. However, both palaeomagnetic and $^{40}Ar/^{39}Ar$ evidence indicate that much of the intrusive igneous activity took place between 230 and 220 Ma, thus post-dating the main pulse of trap volcanism by at least 20 Myr. Only one basaltic flow, at the southern extreme of the studied area, yields an age (248.5 ± 6.0 Ma, 2σ) consistent with trap volcanism. This unit is shown by field relationships to be older than the other dated sills in the area.

The main folding event is constrained by a positive palaeomagnetic fold test to post-date the 230–220 Ma emplacement of the sills, but must pre-date the flat-lying sediments of Hettangian–lower Pliensbachian age that unconformably overly the studied units. An age of 198 Ma, obtained from the chilled margin of one sill, is tentatively interpreted as a maximum age for partial resetting of the argon isotopes in connection with the folding event, and is compatible with deformation around the Triassic–Jurassic boundary.

Different models exist for the collision of the Kara Block with the Siberian margin. Structural and stratigraphic evidence from South-Central Taimyr suggest relatively minor Permo-Carboniferous thin-skinned 'Uralian' collision followed by more intense partitioned dextral transpression in Late Triassic–earliest Jurassic time during which the principal folding occurred (Inger *et al.* 1999). The collective palaeomagnetic and $^{40}Ar/^{39}Ar$ data of this study and the study of Torsvik & Andersen (2002) do not show any trace of the Permo-Carboniferous collision, but clearly document and further constrain the Late Triassic–earliest Jurassic event. Whatever the cause, the Early Mesozoic deformation in South-Central Taimyr is distinct from Late Palaeozoic Uralian age activity, and the occurrence of contemporaneous Early Mesozoic activity elsewhere in Arctic Russia implies a tectonic event of regional significance.

ACKNOWLEDGMENTS

Helpful reviews by Christoph Heunemann and Giovanni Muttoni are gratefully acknowledged. We thank SWEDARP and EUROPROBE for providing very thorough logistical support prior to and during the field season in Taimyr. The fieldwork was financially supported by a grant from the Norwegian Research Council to one of us (HJW), and the subsequent write-up was partially supported by an INTAS award (INTAS-RFBR-97-1139). The CASP contribution was funded by a consortium of oil companies. L. R. M. Cocks and T. H. Torsvik are thanked for stimulating discussion about key aspects of Arctic Siberia palaeogeography and palaeomagnetic data, respectively.

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APPENDIX: ANALYTICAL PROCEDURES, GEOLOGICAL SURVEY OF NORWAY ^{40}AR/^{39}AR GEOCHRONOLOGY LABORATORY; COMPLETE DATA TABLES FOR FURNACE STEP-HEATING EXPERIMENTS

Mineral separates were obtained using standard techniques. Before packing in Al-foil, mineral separates were hand-picked under binocular microscope and then rinsed in alternating acetone and distilled water. Sample packets were stacked and loaded in a sealed Al-capsule with Cd-shielding for irradiation in the 5C site at the McMaster Nuclear Reactor facility, Hamilton, Canada. The samples at McMaster were irradiated for 25 hr (50 MWH) with nominal neutron flux of 4 × 10e13 n cm^{-2} s^{-1}; nominal temperature in the irradiation site is <50°C (M. Butler, personal communication, 2001). Production of isotopes from Ca and K were determined by irradiation of CaF and K_2SO_4 salts; values of 36/37Ca = 0.00028, 39/37Ca = 0.001195, and 40/39K = 0.025 were used. Neutron fluence monitors included Taylor Creek Rhyolite sanidine, Tinto biotite and Hb3gr hornblende with ages of 27.92 Ma (TCR; Duffield & Dalrymple 1990), 410.3 Ma (Tinto; Rex & Guise 1995) and 1072 ± 11 Ma (Hb3gr; Turner *et al.* 1971). Uncertainty in *J*-values for monitors were usually <1 per cent without including error in monitor age; we incorporated a conservative 1 per cent error in *J*-value for all unknowns in the last column of Table 3 (see text for details).

Samples were analysed at the ^{40}Ar/^{39}Ar Geochronology Laboratory at the Norwegian Geological Survey with equipment and analysis procedures as described in Eide *et al.* (2002). Gas from irradiated samples was released in a stepwise fashion from a resistance furnace. Masses 37–40 were each measured in eight cycles with 10 counts per mass per cycle with exception of mass 36 which was measured with 20 counts per cycle. Dynamic blank measurements on mass 40 indicate a stable background (2.19 × 10^{-14} ccSTP), or just detectable above instrument background levels (between-mass signal levels). Background levels (blanks) for the furnace were measured at 100–200°C temperature increments at the start of each sample analysis. Blanks were maintained at levels of 3–9 × 10^{-12} ccSTP for mass 40 and 4–9 × 10^{-14} ccSTP for mass 36 at temperatures of 500–1000°C; blanks increased to 1–6 × 10^{-11} ccSTP for mass 40 and 2 × 10^{-13} ccSTP for mass 36 at high temperatures (1200–1400°C). At experimental temperatures between 600 and 1000°C, furnace blanks for mass 40 were usually <1 per cent of the sample signal size. Background levels of masses 37 and 39 did not change significantly from dynamic blank levels at any temperature (2.1 × 10^{-13} ccSTP for mass 37; <3.3 × 10^{-14} ccSTP for mass 39). Background levels for mass 38 were at dynamic blank levels 1 × 10^{-13} ccSTP at high temperatures and < 9 × 10^{-14} ccSTP at temperatures <1000°C. Data reduction incorporates corrections for interfering isotopes, mass discrimination, error in blanks and decay of ^{37}Ar and ^{39}Ar.

Table A1. Complete ^{40}Ar/^{39}Ar data tables.

Temp (C)	^{40}Ar/^{39}Ar	^{38}Ar/^{39}Ar	^{37}Ar/^{39}Ar	^{36}Ar/^{39}Ar	^{39}Ar (10^{-15} mol)	F^{39}Ar[1]	per cent 40^{*2}	^{40}Ar*/^{39}Ar$_K$	Age (Ma)	±1σ
TAI-10 plagioclase			$J = 0.0061580$							
500	23.572	0.008	1.298	5.937	0.569	3.28	93.1	21.93	227.36	0.90
600	36.001	0.003	2.923	49.765	0.625	6.90	60.0	21.59	223.98	1.33
700	75.468	0.012	5.409	163.996	0.790	11.46	36.6	27.61	281.80	2.54
800	24.356	0.006	3.542	9.063	3.342	30.77	90.5	22.04	228.39	0.67
840	22.845	0.003	1.497	3.835	2.794	46.91	95.6	21.85	226.55	0.32
880	23.22	0.002	1.243	3.042	1.866	57.68	96.6	22.43	232.21	0.57
910	23.438	0.003	1.045	5.489	1.578	66.80	93.5	21.91	227.08	0.28
950	23.356	0.003	0.858	4.715	1.795	77.17	94.3	22.03	228.31	0.47
1000	25.314	0.007	1.224	11.903	1.928	88.30	86.5	21.90	227.08	0.80
1040	24.988	0.005	2.191	10.300	0.855	93.24	88.7	22.16	229.55	0.70
1080	27.962	0.006	4.790	21.048	0.493	96.09	79.5	22.24	230.36	1.31
1150	25.467	0.007	3.945	13.738	0.677	100	85.6	21.81	226.17	0.84
T22 plagioclase			$J = 0.0061880$							
500	75.15	0.032	18.132	170.027	0.139	4.80	35.9	26.97	278.57	4.18
600	80.096	0.015	27.927	183.947	0.274	14.25	36.2	28.99	297.87	6.12
700	114.239	0.021	28.587	297.275	0.424	28.90	26.0	29.75	305.03	7.63
800	67.066	0.019	24.389	149.101	0.284	38.71	38.4	25.75	266.89	4.20
850	28.995	0.000	24.208	32.529	0.178	44.84	75.9	22.00	230.36	3.24
915	31.408	0.007	17.975	34.757	0.191	51.45	73.5	23.10	241.13	3.06
985	31.779	0.000	16.825	35.452	0.182	57.74	72.8	23.14	241.52	2.68
1055	29.325	0.006	14.331	24.915	0.245	66.21	80.2	23.53	245.33	2.11
1125	28.911	0.007	12.222	28.924	0.320	77.24	74.9	21.67	227.11	1.93
1200	28.956	0.005	19.662	34.623	0.357	89.58	71.9	20.82	218.71	2.93
1275	26.315	0.007	16.783	38.424	0.302	100	63.3	16.66	177.10	1.32
TAI-5 plagioclase			$J = 0.0061480$							
500	83.412	0.010	10.662	176.967	0.353	5.63	38.8	32.39	327.78	4.23
600	153.969	0.015	19.312	381.521	0.575	14.80	28.5	43.81	430.55	5.96
700	47.511	0.007	19.460	79.843	1.257	34.86	55.0	26.11	268.72	1.10
750	26.823	0.001	20.729	16.735	0.360	40.59	90.1	24.17	250.05	3.05
800	34.472	0.019	21.986	40.703	0.421	47.31	72.2	24.89	257.04	2.84
875	43.192	0.020	17.525	25.257	0.447	54.45	87.9	37.95	378.52	5.73

Table A1. *(Continued.)*

Temp (C)	$^{40}Ar/^{39}Ar$	$^{38}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	^{39}Ar (10^{-15} mol)	$F^{39}Ar$[1]	per cent 40*[2]	$^{40}Ar^*/^{39}Ar_K$	Age (Ma)	$\pm 1\sigma$
TAI-5 plagioclase			$J = 0.0061480$							
950	32.998	0.008	12.530	38.148	0.445	61.55	70.0	23.08	239.54	2.25
1000	33.757	0.007	12.475	43.630	0.336	66.92	65.8	22.20	230.96	1.75
1040	40.106	0.006	14.453	53.152	0.438	73.91	64.9	26.02	267.86	1.71
1100	43.649	0.008	15.496	57.609	0.544	82.58	65.1	28.41	290.57	1.52
1150	56.333	0.008	17.567	72.524	0.285	87.13	65.9	37.11	370.96	1.83
1200	50.815	0.005	19.209	58.462	0.386	93.28	70.7	35.93	360.25	1.87
1250	56.757	0.002	19.456	57.014	0.225	96.87	74.8	42.48	418.86	4.56
1300	50.68	0.005	18.404	53.647	0.196	100	73.3	37.14	371.25	3.06
TAI-7 plagioclase			$J = 0.0061530$							
500	29.566	0.003	10.225	45.493	0.364	5.37	58.0	17.15	181.12	2.34
600	55.465	0.008	20.84	118.338	0.704	15.76	41.0	22.76	236.61	3.75
700	84.600	0.013	17.836	202.391	0.789	27.41	31.7	26.82	275.67	3.49
800	35.459	0.007	10.128	50.265	0.939	41.27	61.1	21.68	226.04	1.33
975	20.995	0.003	3.402	7.304	1.085	57.29	91.3	19.17	201.28	0.59
1035	20.610	0.004	3.192	6.705	1.328	76.90	91.9	18.94	198.98	0.47
1100	21.127	0.005	4.944	7.843	0.742	87.86	91.4	19.31	202.63	0.83
1155	23.140	0.004	10.903	15.99	0.395	93.68	84.5	19.55	205.00	1.21
1230	25.426	0.004	24.036	21.839	0.428	100	84.8	21.56	224.81	1.97
TAI-8 plagioclase			$J = 0.0061580$							
500	41.375	0.017	50.869	81.121	0.054	3.86	55.6	22.99	238.97	16.02
600	240.303	0.047	60.769	704.629	0.107	11.50	16.6	39.99	397.42	16.53
700	100.132	0.019	77.762	261.400	0.202	25.88	32.3	32.30	327.46	4.94
750	29.614	0.012	75.123	45.754	0.138	35.69	82.7	24.48	253.50	3.39
800	42.567	0.000	71.537	85.660	0.128	44.79	59.5	25.31	261.49	4.17
850	31.784	0.003	54.479	52.702	0.124	53.63	69.6	22.14	230.66	3.55
900	36.355	0.000	50.248	51.699	0.091	60.09	73.8	26.82	275.93	6.42
950	38.906	0.000	52.761	43.922	0.067	64.84	83.0	32.31	327.51	7.89
1000	35.094	0.000	66.780	57.890	0.106	72.35	72.7	25.53	263.54	4.33
1050	38.995	0.000	62.583	69.848	0.085	78.38	65.2	25.41	262.42	4.26
1100	33.237	0.005	66.462	55.416	0.068	83.21	73.0	24.27	251.38	6.50
1150	24.588	0.006	54.636	23.396	0.102	90.50	96.5	23.72	246.10	3.16
1200	24.463	0.007	48.437	18.674	0.134	100	99.5	24.34	252.07	3.25

(1) Cumulative ^{39}Ar gas released during the experiment (per cent).
(2) per cent 40^* = per cent radiogenic ^{40}Ar of total ^{40}Ar released.